Lesson Planning

Complete Lesson Plan Book

Class: Class 8

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Computer Networks

Learning Objectives

• Define a computer network.
• Distinguish between LAN, MAN and WAN.
• Explain the uses of networks.
• Identify network devices: hub, switch, router.

Key Concepts

• A network connects computers to share data and resources.
• LAN covers a small area like a school; WAN covers countries.
• The internet is the largest WAN.
• A router connects networks together.
• Networks save cost by sharing printers and internet.

Teaching Methodology (Step by Step)

1. Ask how the lab computers share one printer.
2. Define network and draw a small LAN diagram.
3. Compare LAN, MAN and WAN with examples.
4. Introduce hub, switch and router.
5. Students draw the school lab network.

Classroom Activities

• Draw and label a LAN diagram.
• Identify network devices in the lab.

Assessment Questions

1. Define a computer network.
2. Distinguish between LAN and WAN.
3. What is the function of a router?

Homework

• Draw the network you would set up at home.

Time Allocation

Discussion 5 min, teaching 20 min, drawing 10 min, quiz 5 min.

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Unit 2: MS Excel: Formulas and Functions

Learning Objectives

• Use arithmetic formulas in Excel.

• Apply SUM, AVERAGE, MAX, MIN and COUNT.

• Use the IF function for decisions.

• Sort and filter data.

Key Concepts

• Formulas start with = and use cell addresses.

• Functions are ready-made formulas.

• =IF(A1>=40,"Pass","Fail") makes decisions.

• Sorting arranges data; filtering shows selected rows.

• Absolute references (A1) don't change when copied.

Teaching Methodology (Step by Step)

1. Revise basic Excel from class 7.
2. Demonstrate SUM, AVERAGE, MAX, MIN, COUNT on marks data.
3. Teach the IF function with pass/fail.
4. Show sorting by total marks.
5. Show filtering to see only passed students.

Classroom Activities

• Build a full result card with formulas.

• Sort and filter the class data.

Assessment Questions

1. Write the formula to find the average of B2:B11.
2. Write an IF formula for pass/fail at 40 marks.
3. What is the difference between sorting and filtering?

Homework

• Make a family budget sheet with SUM formulas.

Time Allocation

Revision 5 min, demo 20 min, practice 15 min.

Unit 3: MS Excel: Charts

Learning Objectives

• Explain the purpose of charts.

• Create column, bar and pie charts.

• Choose the right chart for data.

• Format charts with titles and labels.

Key Concepts

• Charts show data visually.

• Column charts compare values; pie charts show parts of a whole.

• Every chart needs a title and labelled axes.

• Data labels make charts easy to read.

• Wrong chart types can mislead.

Teaching Methodology (Step by Step)

1. Show the same marks data as a table and as a chart — compare.
2. Create a column chart step by step.
3. Create a pie chart of favourite games.
4. Add titles, labels and colours.
5. Students choose the best chart for given data.

Classroom Activities

• Chart the class survey data.

• Chart-type quiz: which chart fits?

Assessment Questions

1. Which chart shows parts of a whole?
2. Write the steps to create a column chart.
3. Why do charts need titles?

Homework

• Make a pie chart of your weekly time use.

Time Allocation

Comparison 5 min, demo 20 min, practice 15 min.

Unit 4: Introduction to Databases

Learning Objectives

• Define a database.

• Explain tables, records and fields.

• Distinguish between flat and relational databases.

• Give examples of databases in daily life.

Key Concepts

• A database is organised data stored electronically.

• A table has rows (records) and columns (fields).

• Each record describes one item, e.g. one student.

• The key field uniquely identifies each record.

• Examples: school records, bank accounts, NADRA.

Teaching Methodology (Step by Step)

1. Show the attendance register as a paper database.
2. Define database, table, record and field.
3. Design a student table on the board.
4. Explain the key field with roll numbers.
5. Students design a library book table.

Classroom Activities

• Design a database table for the class.

• Field-sorting card activity.

Assessment Questions

1. Define database, record and field.
2. What is a key field? Give an example.
3. Name two databases used in daily life.

Homework

• Design a table for your home library.

Time Allocation

Analogy 5 min, teaching 20 min, design 10 min, review 5 min.

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Unit 5: Introduction to Programming

Learning Objectives

• Define a program and programming language.

• Explain algorithms with daily-life examples.

• Draw simple flowcharts.

• Write a simple program in Scratch/Python style steps.

Key Concepts

• A program is a set of instructions for the computer.

• An algorithm is a step-by-step solution.

• Flowcharts use symbols: oval (start/end), rectangle (process), diamond (decision).

• Languages: Python, Scratch, C++.

• Programs must be logical and ordered.

Teaching Methodology (Step by Step)

1. Write the algorithm for making tea.
2. Define program, language and algorithm.
3. Introduce flowchart symbols.
4. Draw a flowchart for finding the larger of two numbers.
5. Students write an algorithm for brushing teeth.

Classroom Activities

• Flowchart drawing competition.

• Algorithm treasure hunt.

Assessment Questions

1. Define algorithm and program.
2. Draw the three main flowchart symbols.
3. Write the algorithm to add two numbers.

Homework

• Draw a flowchart for morning routine decisions.

Time Allocation

Activity 5 min, teaching 20 min, drawing 10 min, review 5 min.

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Unit 6: Internet Services: Beyond Browsing

Learning Objectives

• Explain online forms and e-commerce.

• Describe video conferencing.

• Explain cloud storage.

• State the rules of digital citizenship.

Key Concepts

• E-commerce means buying and selling online.

• Video conferencing connects people face to face.

• Cloud storage (Google Drive) keeps files online.

• Digital citizens are respectful and honest online.

• Think before you post or share.

Teaching Methodology (Step by Step)

1. Discuss online shopping experiences.
2. Demonstrate filling an online form safely.
3. Demonstrate a video call setup.
4. Show saving a file to Google Drive.
5. Teach digital citizenship rules.

Classroom Activities

• Mock online order form filling.

• Digital citizenship pledge.

Assessment Questions

1. What is e-commerce?
2. Name two uses of cloud storage.
3. Write three rules of digital citizenship.

Homework

• List five internet services with one use each.

Time Allocation

Discussion 5 min, demos 20 min, pledge 10 min, review 5 min.

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Unit 7: Multimedia Basics

Learning Objectives

• Define multimedia and its elements.

• Explain the uses of images, audio and video.

• Create a simple presentation with media.

• State the rules of using others' media (copyright).

Key Concepts

• Multimedia combines text, images, audio, video and animation.

• Images and video make ideas clear.

• Presentations are made in PowerPoint.

• Copyright protects creators' work.

• Always credit the source of media.

Teaching Methodology (Step by Step)

1. Show a plain slide vs a multimedia slide — compare.
2. Introduce the five elements of multimedia.
3. Demonstrate inserting pictures, audio and video in PowerPoint.
4. Show transitions and animations.
5. Teach copyright and crediting.

Classroom Activities

• Create a 5-slide multimedia presentation.

• Best presentation showcase.

Assessment Questions

1. Name the five elements of multimedia.
2. What is copyright?
3. Why should you credit media sources?

Homework

• Make a presentation on 'My Country'.

Time Allocation

Comparison 5 min, demo 20 min, creation 15 min.

Unit 8: Cyber Safety and Digital Ethics

Learning Objectives

• Define cybercrime.

• Explain password safety.

• Identify phishing and online scams.

• Practise responsible social media use.

Key Concepts

• Cybercrime is crime committed using computers.

• Strong passwords mix letters, numbers and symbols.

• Phishing tricks you into sharing secrets.

• Never meet online strangers alone.

• Report abuse to trusted adults.

Teaching Methodology (Step by Step)

1. Share a story of an online scam.
2. Teach strong-password rules.
3. Show examples of phishing messages.
4. Discuss social media privacy settings.
5. Students create a personal safety checklist.

Classroom Activities

• Password strength tester activity.

• Spot-the-scam quiz.

Assessment Questions

1. What makes a strong password?
2. What is phishing?
3. Write three rules of safe social media use.

Homework

• Teach your family three cyber safety rules.

Time Allocation

Story 5 min, teaching 20 min, quiz 10 min, checklist 5 min.

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